

6.5 Static Member Functions

This type of variable is declared using the keyword `static` followed by the data type and name of the variable. This is shown in the program given below. The static member variable has some special features. They are initialized with 0 when the first object is created. Only one copy of the variable is created, irrespective of the number of objects declared. The same copy is shared by all the objects. A member function is called static if it has the following properties:

- While calling a static member function, the class name should be used instead of its objects:

```
class-name :: function-name
```

- A function declared static can have access to only other static members - functions and variables declared within the same class.

This is highlighted in the following program.

```
#include <iostream.h>
class t1
{
    int x;
    static int y; // This is a static member variable
public:
    void set(void)
    {
        x= ++y;
    }
    void show(void)
    {
        cout<< "object number: " <<x <<"\n";
    }
}
```